

Chapter 5:

Minority Carrier Diffusion Equation Review
Finish Quasi-Fermi Levels

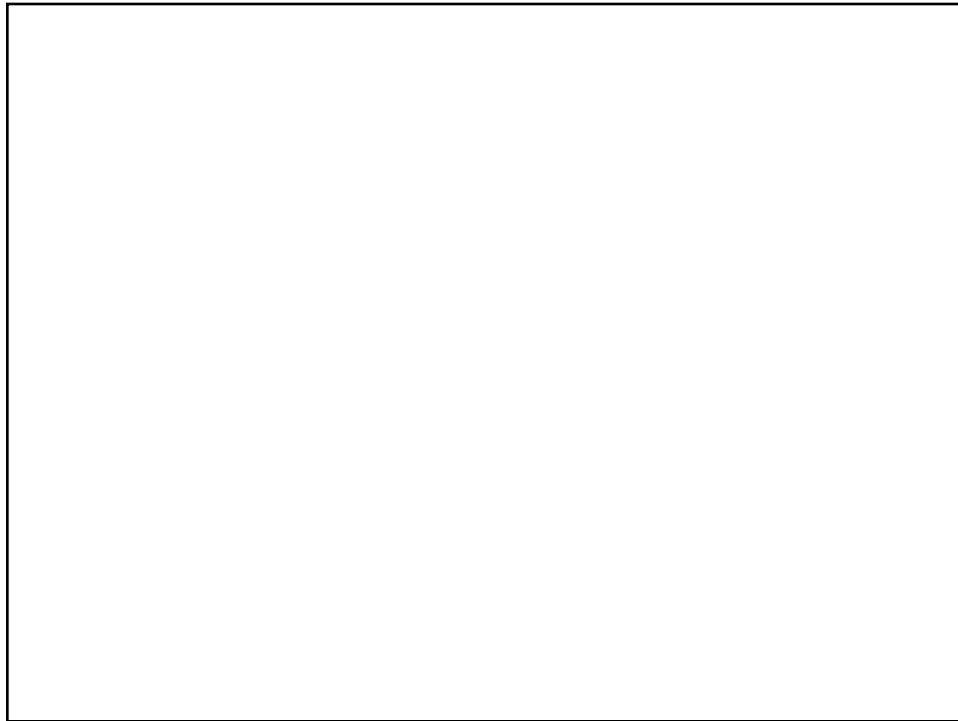
PN Junctions in Equilibrium

Streetman and Banerjee, 'Solid State Electronic
Devices,' 6th ed., Pearson, 2006.

1

Questions?

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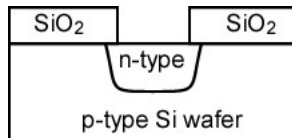
Ch. 5 PN Junctions

1. Equilibrium Band Diagram – Know how to draw the band diagram.
2. Electrostatics - Understand charge and electrostatic potential throughout the pn junction.
 1. Depletion Approximation
 2. Charge
 3. Electric Field
 4. Potential
 5. Built-in voltage
 6. Depletion Width
 7. Capacitance of the junction
3. Current
 1. Use MCDEs to derive the Ideal Diode Equation
 2. Non-idealities

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Making a PN Junction

- At right, fabrication steps of a pn junction.
- Close-up of pn junction shown below.



- Make the n-region by
 - Thermal Diffusion
 - Deposit and thermal drive in.
 - Graded junction
 - Ion Implantation
 - Implant dopant ions at high energy (keV). Thermal anneal to repair damage. Need substitutional doping. Anneal allows dopant atoms to find their spots in the lattice.

1. Oxidize the Si sample

2. Apply a layer of positive photoresist (PR)

3. Expose PR through mask A

4. Remove exposed PR

5. Use RIE to remove SiO₂ in windows

6. Implant boron through windows in the PR and SiO₂ layers

7. Remove PR and sputter Al onto the surface

8. Using PR and mask B, repeat steps 2-4; etch away Al except in p-contact areas

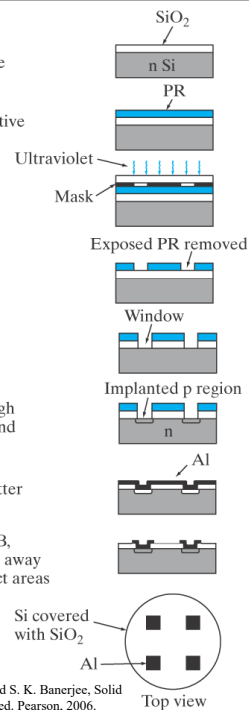
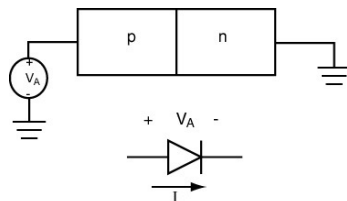


Fig. 5.10. B. G. Streetman and S. K. Banerjee, Solid State Electronic Devices, 6th ed. Pearson, 2006.

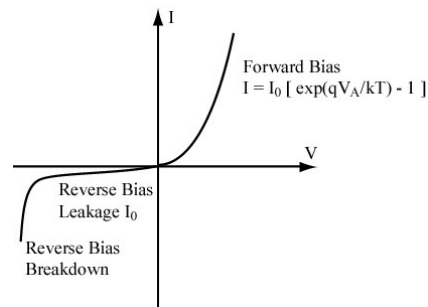
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Idealized PN Junction – EE 100A

- For analysis, the pn junction is schematically represented like this.



I-V Curve



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Drawing the Band Diagram of a PN Junction

PN Junction in **Equilibrium**

- Draw E_c , E_v , E_i , and E_F .



What is the first step?

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Drawing the Band Diagram of a PN Junction

PN Junction in **Equilibrium**

- Draw E_c , E_v , E_i , and E_F .



Step 1. Draw the Fermi Level.

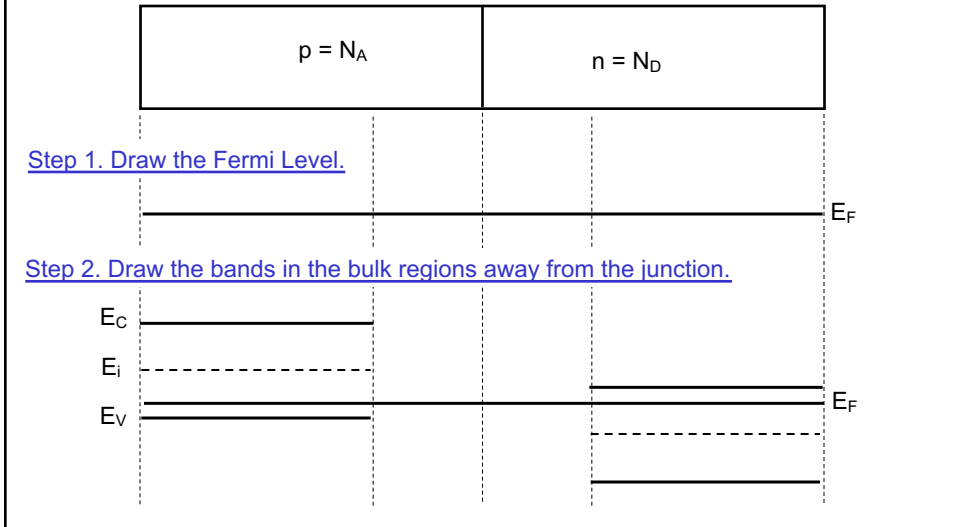
_____ E_F

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Drawing the Band Diagram of a PN Junction

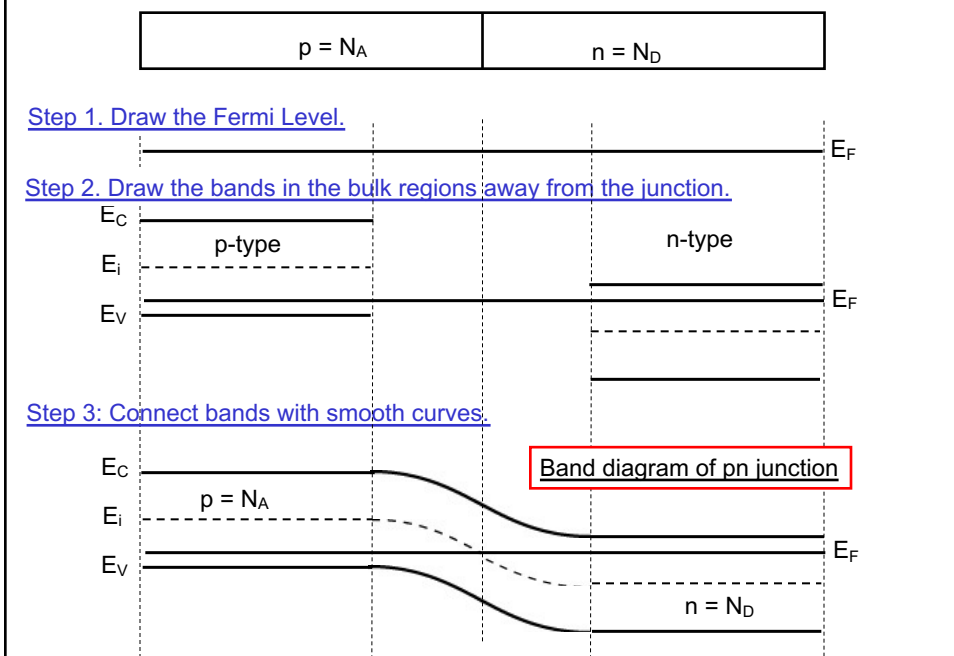
PN Junction in Equilibrium

- Draw E_c , E_v , E_i , and E_F .



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Drawing the Band Diagram (cont.)

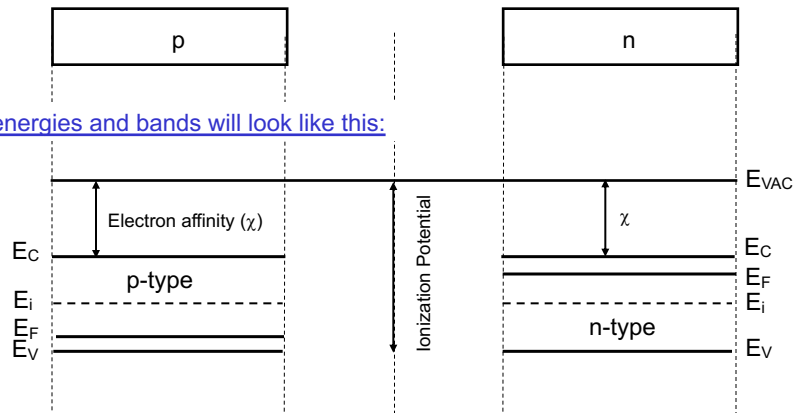


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Contact Potential

Consider n & p type semiconductors that are far apart.

The energies and bands will look like this:

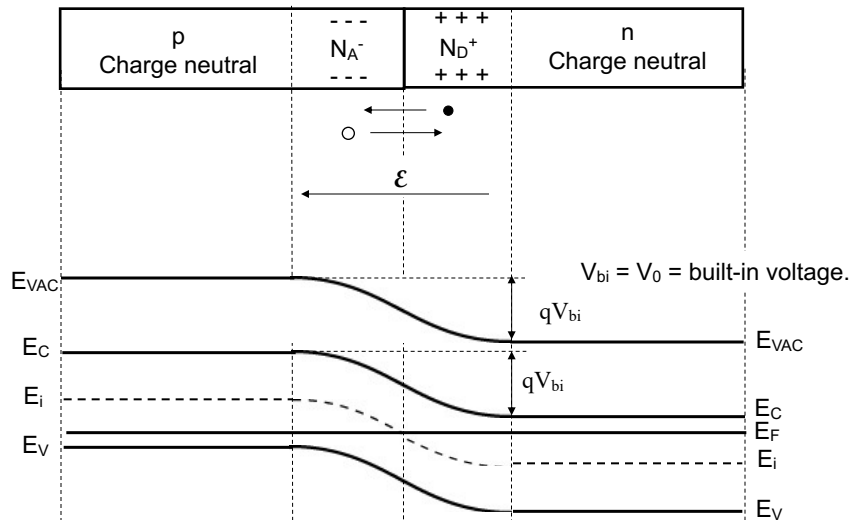


- The reference energy is the vacuum energy.
- The distance from the valence band to the vacuum is the ionization potential.
- The distance from the conduction band to the vacuum is the electron affinity (χ).
- These distances are the same for both pieces of semiconductor since they are both Si.

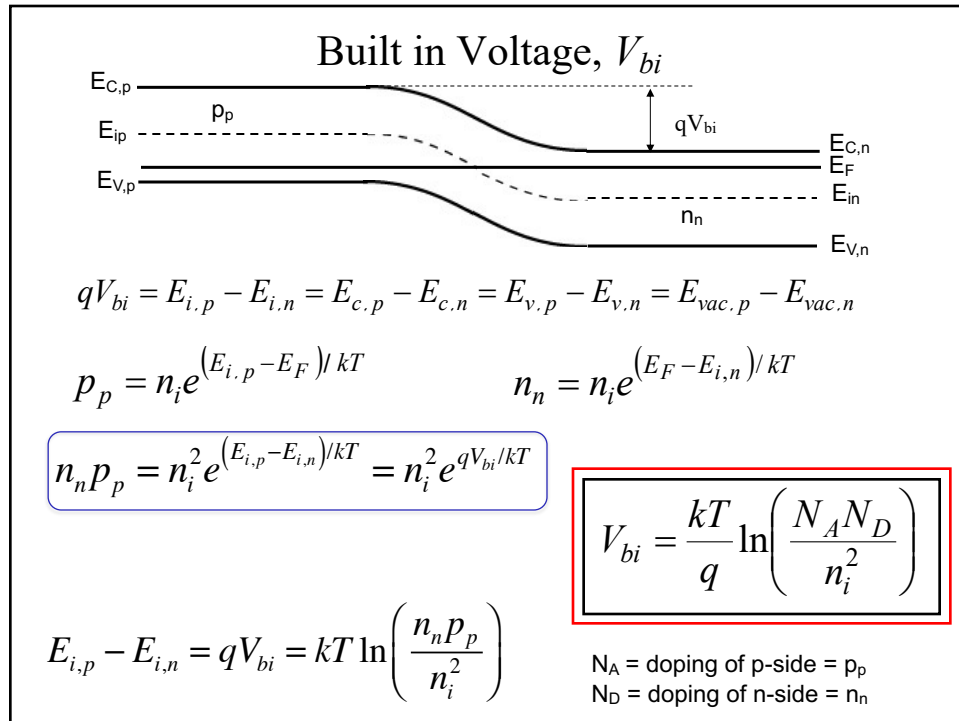
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Contact Potential

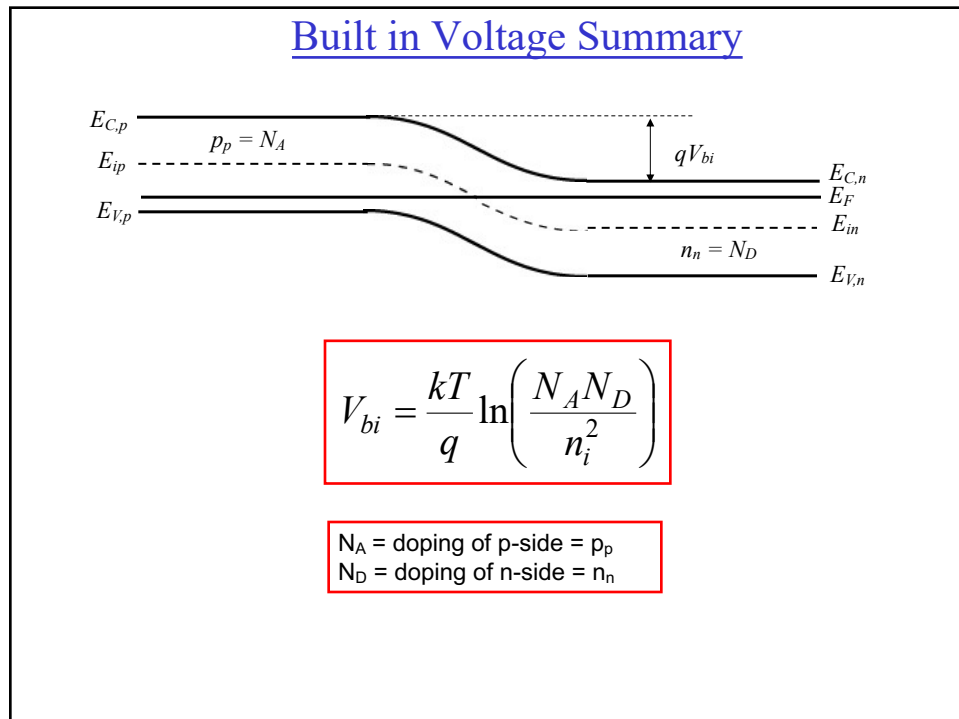
Now bring them into intimate contact.



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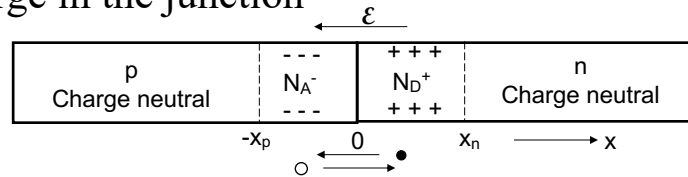
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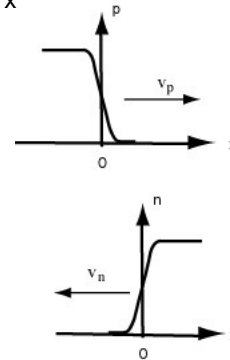
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Electrostatics

• Charge in the junction

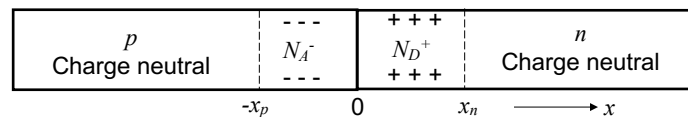


- Electrons and holes diffuse across the junction.
- As the electrons and holes diffuse across the junction, the B^- and P^+ dopants become 'uncovered.' In the junction area, there is no longer $1e^- / P^+$ or $1h^+ / B^-$.
- Unscreened charge N_A^- and N_D^+ in the junction.
- The charge dipole gives rise to an electric field which counteracts the diffusion such that the total current in equilibrium is zero.
- The interface of the n and p regions is referred to as the "metallurgical junction".



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Electrostatics



• Depletion Approximation

- In the depletion region, $-x_p < x < x_n$, there are NO electrons or holes.

- Depletion Width: $W = x_p + x_n$.

The dopants are still there. Therefore, from the charge neutrality equation,

$$\rho = -qn + qp + qN_D^+ - qN_A^-$$

the charge in the junction region is

$$\rho(x) = \begin{cases} -qN_A & (-x_p \leq x < 0) \\ +qN_D & (0 < x \leq x_n) \end{cases} \quad \text{Fixed Charge}$$

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Electrostatics to Get W , x_n , & x_p

- Knowns** are N_A , N_D , V_{bi} , and the form of the charge distribution (from the depletion approximation).

- Unknowns** are x_p and x_n and $W = x_p + x_n$.

- We need 2 equations for 2 unknowns:

1. Charge neutrality $\rightarrow N_A x_p = N_D x_n$

2. Get V_{bi} in terms of x_n and x_p .

- To get V_{bi} , we must integrate the charge twice.

- 1st to get electric field:

$$\frac{d\mathcal{E}}{dx} = \frac{\rho(x)}{\epsilon_s} \quad \text{or, in integral form,} \quad \mathcal{E}(x) = \frac{1}{\epsilon_s} \int_{-x_p}^x \rho(x') dx'$$

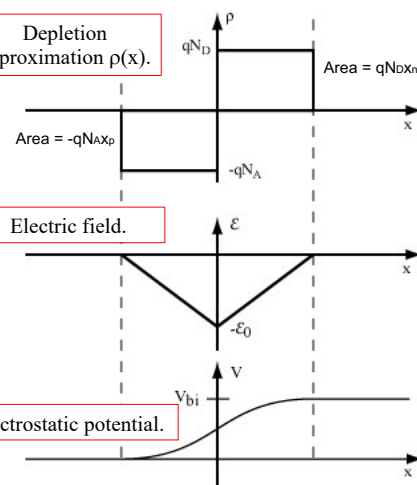
- 2nd to get voltage:

$$\mathcal{E}(x) = -\frac{dV}{dx} \Rightarrow V(x) = -\int_{-x_p}^x \mathcal{E}(x') dx'$$

Charge Neutrality

$$N_A x_p = N_D x_n$$

$p = N_A$ $\rho = 0$	$\rho = -qN_A^-$	$\rho = +qN_D^+$	$n = N_D$ $\rho = 0$
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Electrostatic Analysis

$$V_{bi} = V(x_n) = -\int_{-x_p}^{x_n} dx' \mathcal{E}(x') = \frac{1}{2} \mathcal{E}_0 W$$

$$\mathcal{E}(0) \equiv -\mathcal{E}_0 = \frac{1}{\epsilon_s} \int_{-x_p}^0 dx' (-qN_A) = \frac{-qN_A x_p}{\epsilon_s}$$

$$\mathcal{E}(0) \equiv -\mathcal{E}_0 = \frac{-qN_A x_p}{\epsilon_s} = \frac{-qN_D x_n}{\epsilon_s} \quad \text{Charge neutrality:}$$

$$V_{bi} = \frac{1}{2} \mathcal{E}_0 W = \frac{1}{2} \frac{qN_D}{\epsilon_s} x_n W$$

Use charge neutrality again to get x_n in terms of W :

$$N_A x_p = N_D x_n \Rightarrow x_p = \frac{N_D}{N_A} x_n$$

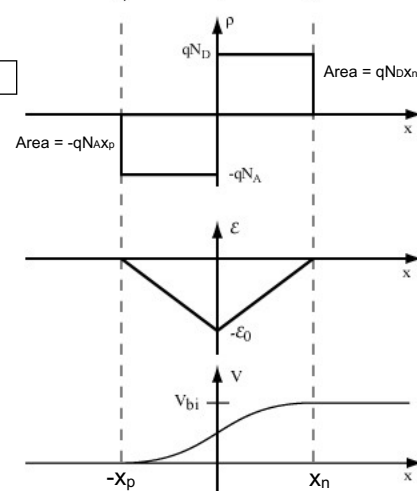
$$W = x_n + x_p = x_n \left(1 + \frac{N_D}{N_A} \right) = x_n \frac{N_A + N_D}{N_A}$$

$$x_n = W \frac{N_A}{N_A + N_D}$$

Charge Neutrality

$$N_A x_p = N_D x_n$$

$p = N_A$ $\rho = 0$	$\rho = -qN_A^-$	$\rho = +qN_D^+$	$n = N_D$ $\rho = 0$
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Electrostatic Analysis

$$V_{bi} = \frac{1}{2} \epsilon_0 W = \frac{1}{2} \frac{q N_D}{\epsilon_s} x_n W$$

$$x_n = W \frac{N_A}{N_A + N_D}$$

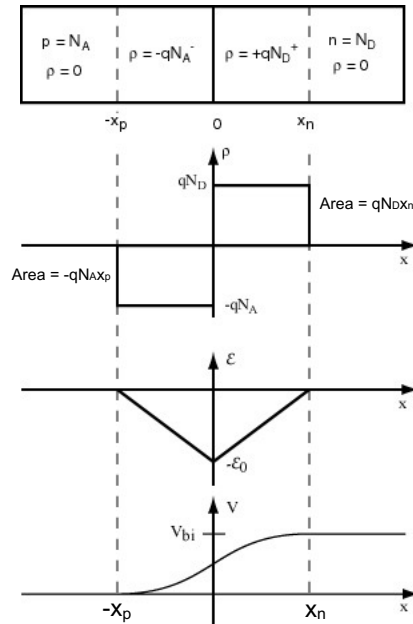
$$V_{bi} = \frac{q}{2\epsilon_s} \frac{N_A N_D}{N_A + N_D} W^2$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \frac{N_A + N_D}{N_A N_D}} \quad V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

ϵ_s is the dielectric constant of the semiconductor. For Si, it is $11.8 \epsilon_0$.

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Often one concentration, say N_D , will be $\gg$ than the other, N_A . Then $1/N_D \ll 1/N_A$, and W is determined by the doping of the p-side.



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Equilibrium Depletion Widths

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

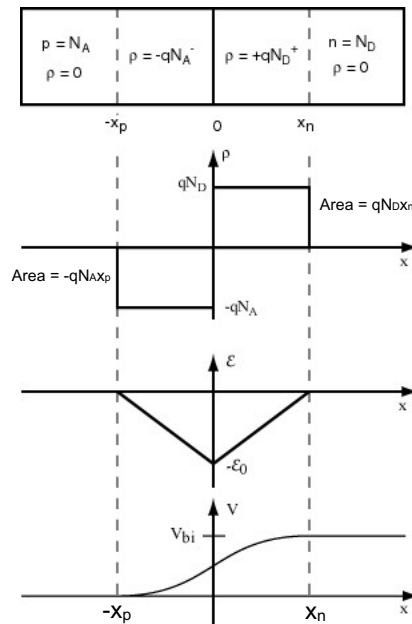
$$x_n = W \frac{N_A}{N_A + N_D} = W \frac{1}{1 + \frac{N_D}{N_A}}$$

$$x_p = W \frac{N_D}{N_A + N_D} = W \frac{1}{1 + \frac{N_A}{N_D}}$$

$$x_n = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \frac{N_A}{N_D (N_A + N_D)}}$$

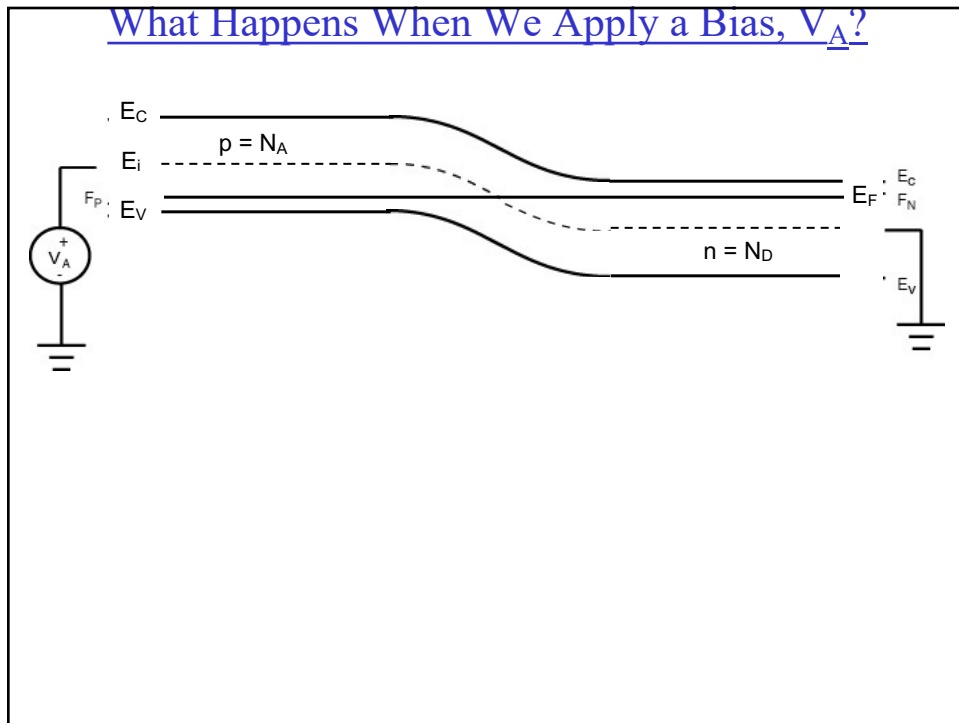
$$x_p = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \frac{N_D}{N_A (N_A + N_D)}}$$

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$



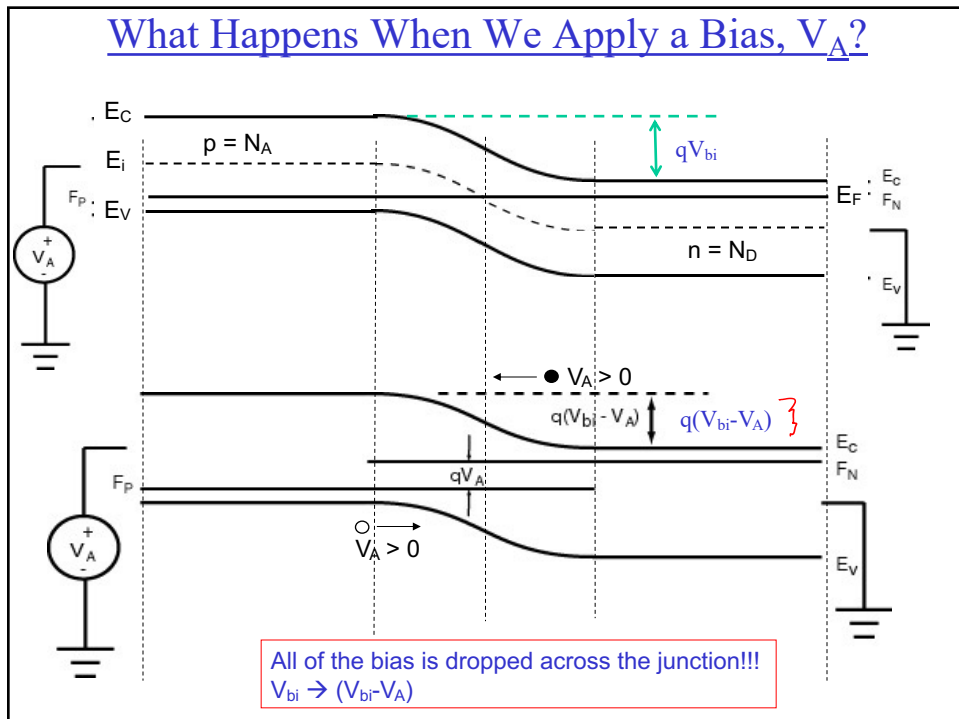
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What Happens When We Apply a Bias, V_A ?



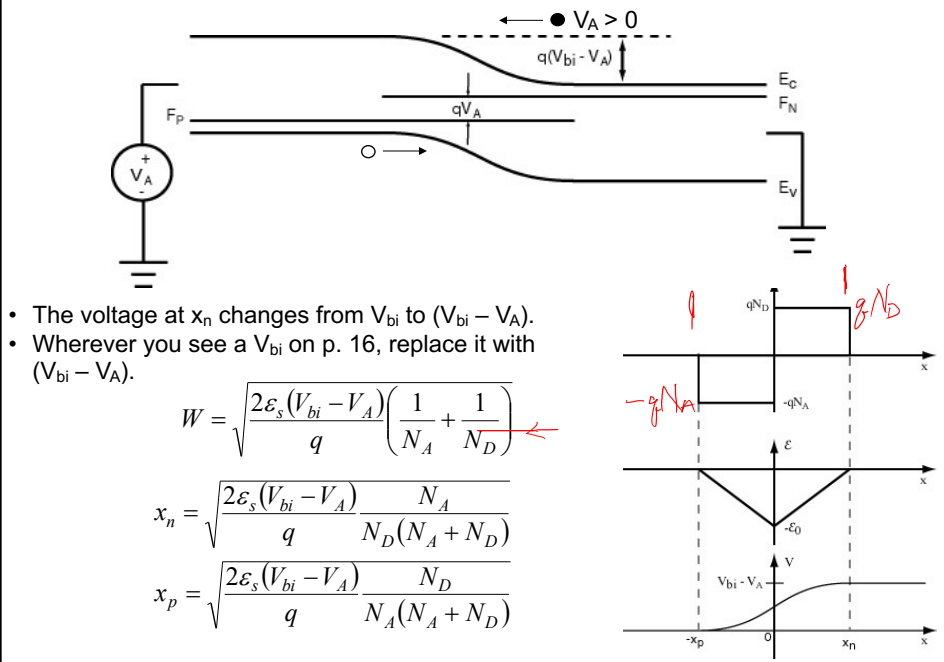
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What Happens When We Apply a Bias, V_A ?



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What Happens When We Apply a Bias, V_A ?



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Non-Equilibrium Depletion Width Summary

With applied
voltage, V_A .

$$W = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)} = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \frac{N_A}{N_D(N_A + N_D)}}$$

$$x_p = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \frac{N_D}{N_A(N_A + N_D)}}$$

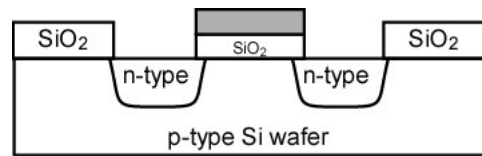
$$x_p = W \frac{N_D}{N_A + N_D} = W \frac{1}{1 + \frac{N_A}{N_D}}$$

$$x_n = W \frac{N_A}{N_A + N_D} = W \frac{1}{1 + \frac{N_D}{N_A}}$$

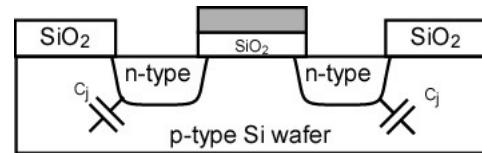
$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

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Junction Capacitance



N-channel FET



FETs, BJTs, and diodes are built upon pn junctions. Each junction contributes a junction capacitance to the circuit. This needs to be understood to model the high frequency or high clock speed performance.

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Junction Capacitance (C_j)

- Referred to as depletion capacitance or junction capacitance. $C_{j1} = \frac{\epsilon_s A}{d}$ $d \rightarrow W$
- Intuitively, we said that $C_j = \epsilon_s / W$ (F/cm²) where ϵ_s is the dielectric constant of the semiconductor and W is the depletion width. It looks just like a parallel plate capacitor with a dielectric or thickness W .

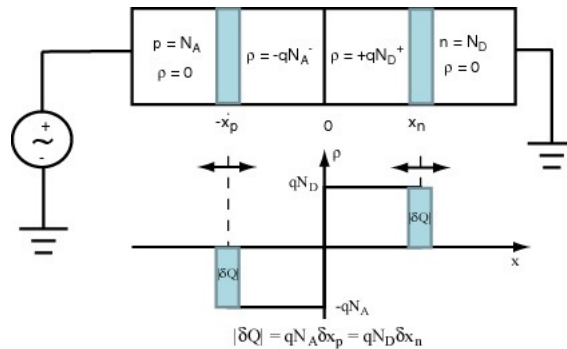
$$C_j = \frac{\epsilon_s}{\sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q}} \left(\frac{N_A + N_D}{N_A N_D} \right)}$$

Valid for negative bias ($V_A < 0$), and V_A slightly positive < turn-on voltage

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Junction Capacitance (C_j)

- Physically, the capacitance arises from the change in charge in the depletion region with change in applied bias.
- The depletion width increases and decreases with bias.



- The change in charge per change in bias defines the capacitance:

$$C_j \equiv \left| \frac{dQ}{dV} \right|$$

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Derive C_j

$$C_j \equiv \left| \frac{dQ}{dV} \right|$$

$$|Q| = qx_n N_D = qx_p N_A = q \frac{N_A N_D}{N_A + N_D} W$$

$$\left| \frac{dQ}{dV_A} \right| = \left| \frac{dQ}{dW} \frac{dW}{dV_A} \right| = q \frac{N_A N_D}{N_A + N_D} \left| \frac{dW}{dV_A} \right|$$

$$q \frac{N_A N_D}{N_A + N_D} \left| \frac{dW}{dV_A} \right| = q \frac{N_A N_D}{N_A + N_D} \left| \frac{dW}{d(V_{bi} - V_A)} \right|$$

$$C_j = q \frac{N_A N_D}{N_A + N_D} \left(\frac{1}{2} \right) \frac{1}{\sqrt{\frac{2\epsilon_s (V_{bi} - V_A)}{q} \frac{N_A + N_D}{N_A N_D}}} \frac{2\epsilon_s (N_A + N_D)}{q N_A N_D}$$

$$C_j = \frac{\epsilon_s}{W}$$

$$x_n = \frac{N_A}{N_A + N_D} W$$

$$x_p = \frac{N_D}{N_A + N_D} W$$

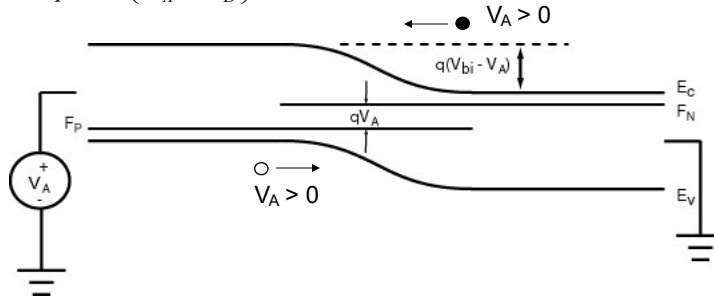
$$W = \sqrt{\frac{2\epsilon_s (V_{bi} - V_A)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)}$$

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Properties of C_j

$$C_j = \frac{\epsilon_s}{\sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}} = \frac{\epsilon_s}{W}$$

Valid for negative bias ($V_A < 0$), and V_A slightly positive $\ll$ turn-on voltage



1. As V_A decreases, W increases and C_j decreases.
2. As N_A or N_D increase, W decreases and C_j increases.
3. For p^+ - n junction, C_j is proportional to $N_D^{1/2}$.
4. For linear graded junction, W proportional to $(V_{bi} - V_A)^{1/3}$.

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Standard Notation for C_j

$$C_j = \frac{\epsilon_s}{\sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)}} = \frac{\epsilon_s}{\left[\frac{2\epsilon_s V_{bi}}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \right]^{1/2} \left(1 - \frac{V_A}{V_{bi}} \right)^{1/2}}$$

$$C_{j0} \equiv \frac{\epsilon_s}{\left[\frac{2\epsilon_s V_{bi}}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \right]^{1/2}}$$

$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}} \right)^{1/2}}$$

In general:
$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}} \right)^m} \quad \begin{matrix} (1/3 < m < 1/2) \\ (-V_{BR} < V_A \leq 0.3V) \end{matrix}$$

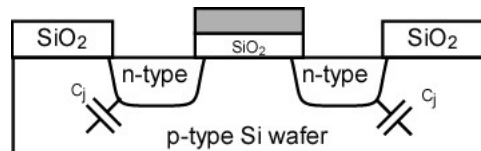
$m = 1/3$ corresponds to a linearly graded junction and $m=1/2$ corresponds to an abrupt junction. ($-V_{BR}$) is the reverse bias breakdown voltage.

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Depletion or Junction Capacitance, C_j

- **SPICE model**
$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}}\right)^m} \quad \begin{matrix} (1/3 < m < 1/2) \\ (-V_{BR} < V_A \leq 0.3V) \end{matrix}$$

- Linearly graded junction means that in the junction region, $N_D - N_A = \alpha x$.
- Abrupt means that N_D and N_A are step functions ending precisely at the metallurgical junction.
- In reality, the profiles are somewhere in between, often with exponential or Gaussian tails extending across the junction.
- Thus, the value for m lies somewhere between the two extremes of abrupt and linearly graded.
- For analysis of linearly graded junction see Sec. 5.2.5 of Pierret.
- Applied wherever there is a pn junction.



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Depletion or Junction Capacitance, C_j

$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}}\right)^m} \quad \begin{matrix} (1/3 < m < 1/2) \\ (-V_{BR} < V_A \leq 0.3V) \end{matrix}$$

- **Varactor** (variable reactor)
 - C_j is a voltage dependent capacitance
 - FM tuners and (old) TVs (tuning $\omega_r = 1/\sqrt{LC}$)
 - Reverse biased pn diode. Change $V_A \rightarrow$ change C_j .

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Depletion or Junction Capacitance, C_j

- How fast can this capacitor respond to a change in voltage?
 - dQ is the result of a movement of **majority** carriers at the edge of the junction.
 - Extremely fast response – dielectric relaxation time.
 - Write the continuity equation for majority carriers:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot J = 0 \quad \text{and} \quad J = \sigma \mathcal{E}$$

$$\frac{\partial \rho}{\partial t} + \sigma \nabla \cdot \mathcal{E} = 0 \quad \text{and} \quad \nabla \cdot \mathcal{E} = \rho / \epsilon$$

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon} \rho = 0 \Rightarrow \rho = \rho_0 e^{-t/\tau} \quad \text{where} \quad \tau = \frac{\epsilon}{\sigma}$$

- Putting in some generic numbers: $\mu_n = 1000 \text{ cm}^2/\text{Vs}$, $n = 10^{18}/\text{cm}^3$, $\sigma = q\mu_n n = 160 (\Omega\text{cm})^{-1}$, $\epsilon = 12(8.854\text{e-}14)(\text{F/cm}) \Rightarrow \tau = 7 \text{ fs}$. Very fast. Good for high frequency applications.
- Note that ϵ/σ looks like an RC time constant.

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T = 300\text{K}$, $\epsilon_s = 11.8 \times 8.854\text{e-}14 \text{ F/cm}$, $n_i = 1.0\text{e}10 / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854\text{e-14 F/cm}$, $n_i = 1.0\text{e10} / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .
 - What first?

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854\text{e-14 F/cm}$, $n_i = 1.0\text{e10} / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.026 \times \ln \left(\frac{10^{33}}{10^{20}} \right) = 0.7783 \text{ V}$$

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854\text{e-}14 \text{ F/cm}$, $n_i = 1.0\text{e}10 / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.026 \times \ln \left(\frac{10^{33}}{10^{20}} \right) = 0.7783 \text{ V}$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)} = \sqrt{\frac{2(11.9)(8.854\text{e-}14)(0.7783)}{1.602\text{e-}19} \left(\frac{1}{10^{18}} + \frac{1}{10^{15}} \right)} = 1.01 \times 10^{-4} \text{ cm}$$

$$W = 1.01 \mu\text{m}$$

Most mistakes occur due to units.

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854\text{e-}14 \text{ F/cm}$, $n_i = 1.0\text{e}10 / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.026 \times \ln \left(\frac{10^{33}}{10^{20}} \right) = 0.7783 \text{ V}$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)} = \sqrt{\frac{2(11.9)(8.854\text{e-}14)(0.7783)}{1.602\text{e-}19} \left(\frac{1}{10^{18}} + \frac{1}{10^{15}} \right)} = 1.01 \times 10^{-4} \text{ cm}$$

$$W = 1.01 \mu\text{m}$$

$$x_n = W \frac{N_A}{N_A + N_D} = 1.01 \times \frac{10^{18}}{10^{18} + 10^{15}} = 1.01 \times \frac{10^{18}}{10^{18}(1 + 0.001)} = \frac{1.01}{1.001} = 1.008991 = 1.009 \mu\text{m}$$

$$x_p = W - x_n = 0.001 \mu\text{m} = 1 \text{ nm}$$

One sided junction

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854 \times 10^{-14} \text{ F/cm}$, $n_i = 1.0 \times 10^{10} / \text{cm}^3$.
 - Calculate W , x_n , x_p and C_{j0} .

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right) = 0.026 \times \ln \left(\frac{10^{33}}{10^{20}} \right) = 0.7783 \text{ V}$$

$$W = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)} = \sqrt{\frac{2(11.9)(8.854 \times 10^{-14})(0.7783)}{1.602 \times 10^{-19}} \left(\frac{1}{10^{18}} + \frac{1}{10^{15}} \right)} = 1.01 \times 10^{-4} \text{ cm}$$

$$W = 1.01 \mu\text{m}$$

$$x_n = W \frac{N_A}{N_A + N_D} = 1.01 \times \frac{10^{18}}{10^{18} + 10^{15}} = 1.01 \times \frac{10^{18}}{10^{18}(1 + 0.001)} = \frac{1.01}{1.001} = 1.008991 = 1.009 \mu\text{m}$$

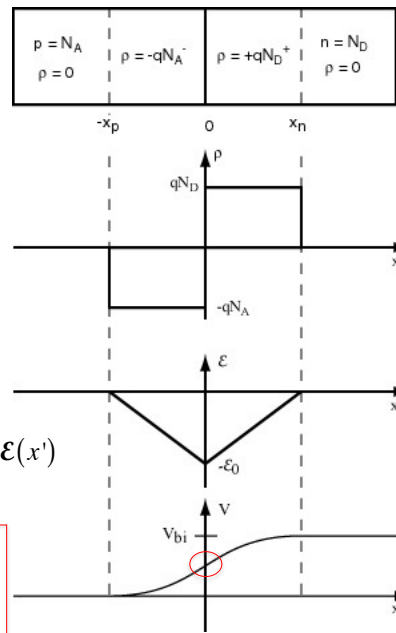
$$x_p = W - x_n = 0.001 \mu\text{m} = 1 \text{ nm}$$

$$C_{j0} = \frac{\epsilon_s}{W} = \frac{11.9(8.854 \times 10^{-14}) \left(\frac{\text{F}}{\text{cm}} \right)}{1.01 \times 10^{-4} \text{ (cm)}} = 1.043 \times 10^{-8} \text{ F/cm}^2 = 10.4 \text{ nF/cm}^2$$

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$, $\epsilon_s = 11.8 \times 8.854 \times 10^{-14} \text{ F/cm}$, $n_i = 1.0 \times 10^{10} / \text{cm}^3$.
 - What is $V(x=0)$?



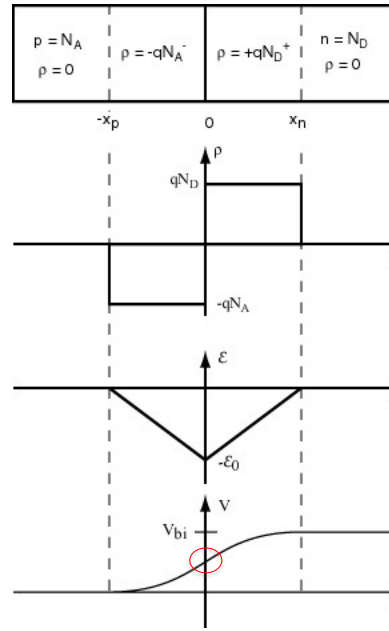
$$-\frac{dV}{dx} = \mathcal{E}(x) \Rightarrow \int_{V(x_p)}^{V(x)} dV = V(x) - V(x_p) = - \int_{-x_p}^x dx' \mathcal{E}(x')$$

$$V(x) = - \int_{-x_p}^x dx' \mathcal{E}(x')$$

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$,
 $\epsilon_s = 11.8 \times 8.854 \times 10^{-14} \text{ F/cm}$, $n_i = 1.0 \times 10^{10} / \text{cm}^3$.
 - What is $V(x=0)$?



$$V(x) = - \int_{-x_p}^x dx' \mathcal{E}(x')$$

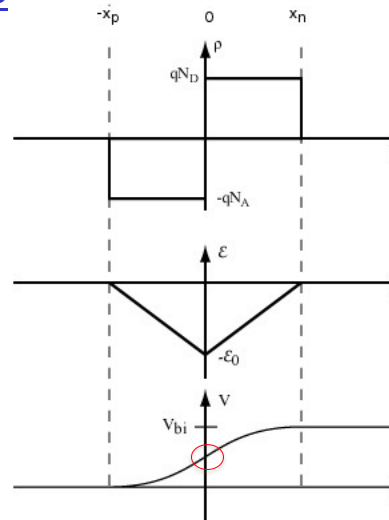
$$\mathcal{E}(0) \equiv \mathcal{E}_0 = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}$$

$$V(x=0) = \frac{1}{2} x_p \mathcal{E}_0 = \frac{1}{2} x_p \frac{qN_A x_p}{\epsilon_s} = \frac{1}{2} \frac{qN_A x_p^2}{\epsilon_s}$$

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$,
 $\epsilon_s = 11.8 \times 8.854 \times 10^{-14} \text{ F/cm}$, $n_i = 1.0 \times 10^{10} / \text{cm}^3$.
 - What is $V(x=0)$?



$$V(x) = - \int_{-x_p}^x dx' \mathcal{E}(x')$$

$$\mathcal{E}(0) \equiv \mathcal{E}_0 = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}$$

$$V(x=0) = \frac{1}{2} x_p \mathcal{E}_0 = \frac{1}{2} x_p \frac{qN_A x_p}{\epsilon_s} = \frac{1}{2} \frac{qN_A x_p^2}{\epsilon_s}$$

$$V(x=0) = \frac{1}{2} \frac{qN_A x_p^2}{\epsilon_s} = \frac{1}{2} \frac{(1.602 \times 10^{-19})(1 \times 10^{18})(1 \times 10^{-7})^2}{11.9(8.854 \times 10^{-14})} = 7.60 \times 10^{-4} \text{ V}$$

$$V(x=0) = 0.76 \text{ mV}$$

One sided junction.

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Example

- Si. $N_A = 10^{18} / \text{cm}^3$, $N_D = 10^{15} / \text{cm}^3$, $T=300\text{K}$,
 $\epsilon_s = 11.8 \times 8.854 \times 10^{-14} \text{ F/cm}$, $n_i = 1.0 \times 10^7 / \text{cm}^3$.
 - What is $V(x=0)$?

$$V(x) = - \int_{-x_p}^x dx' \mathcal{E}(x')$$

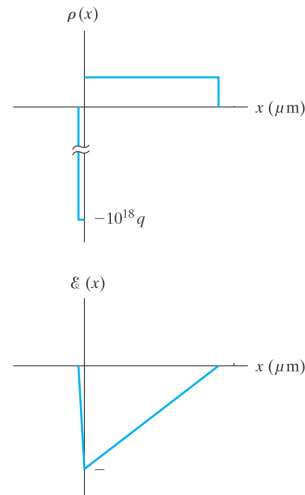
$$-\mathcal{E}(0) \equiv \mathcal{E}_0 = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}$$

$$V(x=0) = \frac{1}{2} x_p \mathcal{E}_0 = \frac{1}{2} x_p \frac{qN_A x_p}{\epsilon_s} = \frac{1}{2} \frac{qN_A x_p^2}{\epsilon_s}$$

$$V(x=0) = \frac{1}{2} \frac{qN_A x_p^2}{\epsilon_s} = \frac{1}{2} \frac{(1.602 \times 10^{-19})(10^{18})(10^{-7})^2}{11.9(8.854 \times 10^{-14})} = 7.60 \times 10^{-4} \text{ V}$$

$$V(x=0) = 0.76 \text{ mV}$$

One sided junction.



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pn-junction depletion width

$$V_{bi} = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

$$W = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)} = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

$$x_n = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \frac{N_A}{N_D(N_A + N_D)}}$$

$$x_p = \sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \frac{N_D}{N_A(N_A + N_D)}}$$

$$-\mathcal{E}(0) \equiv \mathcal{E}_0 = \frac{qN_A x_p}{\epsilon_s} = \frac{qN_D x_n}{\epsilon_s}$$

$$x_p = W \frac{N_D}{N_A + N_D} = W \frac{1}{1 + \frac{N_A}{N_D}}$$

$$x_n = W \frac{N_A}{N_A + N_D} = W \frac{1}{1 + \frac{N_D}{N_A}}$$

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Standard Notation for C_j

$$C_j = \frac{\epsilon_s}{\sqrt{\frac{2\epsilon_s(V_{bi} - V_A)}{q} \left(\frac{N_A + N_D}{N_A N_D} \right)}} = \frac{\epsilon_s}{\left[\frac{2\epsilon_s(V_{bi})}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \right]^{1/2} \left(1 - \frac{V_A}{V_{bi}} \right)^{1/2}}$$

$$C_{j0} \equiv \frac{\epsilon_s}{\left[\frac{2\epsilon_s V_{bi}}{q} \left(\frac{N_A + N_D}{N_A N_D} \right) \right]^{1/2}}$$

$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}} \right)^{1/2}}$$

In general: $C_j = \frac{C_{j0}}{\left(1 - \frac{V_A}{V_{bi}} \right)^m} \quad \begin{matrix} (1/3 < m < 1/2) \\ (-V_{BR} < V_A \leq 0.3V) \end{matrix}$

m = 1/3 corresponds to a linearly graded junction and m=1/2 corresponds to an abrupt junction. (-V_{BR}) is the reverse bias breakdown voltage.